

A Hilbert-space counterexample to the weakly compact thickness equality

Candidate counterexample packet for arXiv:1611.09698

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Status. Candidate full counterexample, likely valid, awaiting specialist review. Question 2 has a negative answer even on a separable infinite-dimensional Hilbert space and for a norm-one weakly compact operator.

1 Source question

Rueda Zoca defines the Daugavet thickness indices

$$\begin{aligned}\mathcal{T}(X) &= \inf \left\{ r > 0 : \begin{array}{l} \text{some nonempty relatively weakly open } W \subset B_X \\ \text{is covered by finitely many } B(x_i, r), \ x_i \in S_X \end{array} \right\}, \\ \mathcal{T}_{w^*}(X^*) &= \inf \left\{ r > 0 : \begin{array}{l} \text{some nonempty relatively weak-star open } W \subset B_{X^*} \\ \text{is covered by finitely many } B(x_i^*, r), \ x_i^* \in S_{X^*} \end{array} \right\}.\end{aligned}$$

Proposition 4.3 gives the lower bound

$$\|T + I\| \geq \max\{\mathcal{T}(X), \mathcal{T}_{w^*}(X^*)\}$$

for norm-one weakly compact T . Question 2 on PDF page 16 then asks whether, for a weakly compact operator $T : X \rightarrow X$,

$$\|T + I\| = \max\{\mathcal{T}(X), \mathcal{T}_{w^*}(X^*)\}. \tag{Q2}$$

See [1].

Moreover, since $w \in T$ we get that $w \in S$. Hence, $X \widehat{\otimes}_\pi Y$ enjoys the Daugavet property, as desired. ■

In view of the previous Proposition and the fact that, up to the best of our knowledge, it is not known whether an ideal in an L -embedded Banach space is itself an L -embedded Banach space (see [24, p. 608]), it is natural to pose the following question.

Question 1. *Let X be an L -embedded space with the Daugavet property and let Y be a non-zero Banach space. If either X^{**} or Y has the metric approximation property, does $X \widehat{\otimes}_\pi Y$ enjoy the Daugavet property?*

It is known that $X \widehat{\otimes}_\pi Y$ has, at least, an octahedral norm under the assumptions of the previous question [22, Theorem 4.3].

With respect to Section 4, in view of the characterisations given in Theorem 2.2 and of Proposition 4.3, it is natural to wonder.

Question 2. *Let X be a Banach space and $T : X \rightarrow X$ be a weakly compact operator. Is it true that $\|T + I\| = \max\{\mathcal{T}(X), \mathcal{T}_{w^*}(X^*)\}$?*

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REFERENCES

Figure 1: Question 2 on PDF page 16 of the source paper.

2 Proof intuition

The right side of (Q2) measures whether some weakly open piece of the unit ball can be covered by small balls. In a Hilbert space, a slice very near a unit vector is an arbitrarily small spherical cap, so both thickness indices vanish. This has no reason to force $\|I + T\|$ to vanish. Taking $T = I$ makes the mismatch maximal while retaining norm one and weak compactness.

3 Counterexample

Theorem 1. *The equality (Q2) is false. It remains false if X is required to be separable and infinite-dimensional and T is required to have norm one.*

Proof. Let $H = \ell_2$ over the real scalars and let $e = e_1$. For $0 < \delta < 1$, consider

$$W_\delta = \{x \in B_H : \langle x, e \rangle > 1 - \delta\}.$$

This is a nonempty relatively weakly open subset of B_H , and $e \in W_\delta$. For every $x \in W_\delta$,

$$\|x - e\|^2 = \|x\|^2 + \|e\|^2 - 2\langle x, e \rangle < 2\delta.$$

Consequently

$$W_\delta \subset B(e, \sqrt{2\delta}) \quad \text{and hence} \quad \mathcal{T}(H) \leq \sqrt{2\delta}.$$

Letting $\delta \downarrow 0$ gives $\mathcal{T}(H) = 0$.

The Hilbert space H is reflexive. Thus the weak-star topology $\sigma(H^*, H)$ and the weak topology $\sigma(H^*, H^{**})$ coincide under the canonical identification $H^{**} = H$. Applying the same cap argument in the Hilbert space H^* gives

$$\mathcal{T}_{w^*}(H^*) = 0.$$

Now take $T = I_H$. It has norm one. It is weakly compact because the closed unit ball of the reflexive space H is weakly compact. But

$$\|T + I_H\| = \|2I_H\| = 2, \quad \max\{\mathcal{T}(H), \mathcal{T}_{w^*}(H^*)\} = 0.$$

This contradicts (Q2). □

4 Scope and novelty

The source's Question 2 omits the norm-one hypothesis present in its immediately preceding proposition. The example above deliberately uses a norm-one operator, so adding that hypothesis does not repair the equality. The same proof works on any nonzero reflexive Banach space whose primal and dual balls have arbitrarily small weak slices; Hilbert space is the cleanest instance.

This packet does not address Question 1 on projective tensor products of L -embedded Daugavet spaces.

A bounded primary-source search through 12 August 2026 used the exact arXiv id and title, the displayed equality, and combinations of *Daugavet thickness*, *weakly compact operator*, and *counterexample*. It found later work on other thickness-index questions but no explicit prior answer to (Q2). Since the contradiction is elementary and the printed question may be an unintended formulation, novelty confidence is moderate; the appropriate claim is a correction to the literal statement.

References

- [1] A. Rueda Zoca, *Daugavet property and separability in Banach spaces*, arXiv:1611.09698 (2016).